

Nicholas McLean

Mechanical Engineer

Contact

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Saskatoon, Canada

Education

BSc in Mechanical Engineering
with Distinction

University of Alberta, 2023

GPA: 3.6 / 4.0

Awards & Scholarships

- Louise McKinney Scholarship, 2021
- Shell Eco-Marathon Americas Vehicle Design Award Runner-Up, 2021
- Jason Lang Scholarship, 2019, 2020
- Vehicle Shop Design Award, 2019
- Alexander Rutherford Scholarship, 2018

Key Skills

- Assembly Automation
- PFMEA
- Traceability & Productivity
- Computer Aided Design (SolidWorks, AutoCAD, Creo)
- Finite Element Analysis (SolidWorks, Ansys)
- Process Development
- Engineering Drawings
- Project Management
- Plant Launches
- Microsoft Office
- Process Optimization

Experience

Tooling Designer

CNH Industrial March 2025 - Present

- Spearheaded tooling development for major Hoe Drill product launch
- Executed scope definition with internal stakeholders, created 3D tooling models and manufacturing drawings using Creo
- Designed custom tooling and fixtures to support production of agricultural products, enhancing manufacturability, and reducing changeover time
- Managed tooling releases via Teamcenter
- Led the RFQ to PO process for external vendors using SAP
- Worked with internal machinists and fixture fabricators to ensure adherence to design intent during the build phase

Process Engineer

Camcor Manufacturing May 2024 - March 2025

- Successfully launched a Rear Drive Unit (RDU) assembly line with over 80 components and 46 operations
- Implemented end-of-line processes including leak testing, NVH and balancing
- Collaborated with design and build teams to align production processes with customer requirements for the assembly of 1 and 2 speed RDU's
- Conducted visits to component suppliers to understand and/or scrutinize processes to improve quality
- Created and maintained standard documentation (operator instructions, process specifications, and visual aids) for semi-automated assembly operations, ensuring clarity and consistency
- Led the development of PFMEA (Process Failure Mode & Effect Analysis) in collaboration with cross functional team.
- Reviewed PFMEA with OEM customer and provided tour of production line

Process Engineer

Linamar Light Metals Welland September 2023 - May 2024

- Played a key role in the strategic planning of a 287,000 sq ft giga casting facility by designing the plant layout for optimal process flow
- Led investigations as a part of the core team to enhance product quality during prototype stages, utilizing MRB, 8D and 5 Why problem-solving methodologies
- Fostered strong partnerships with vendors for the integration of Special Purpose Machines and the advancement of automated systems, ensuring robustness in fixture design and error-proofing (poka-yoke)
- Developed a comprehensive spare parts list and strategically prioritized inventory based on criticality to reduce cost
- Ensured manufacturing processes aligned with client expectations by meticulously verifying adherence to dimensional specifications within customer models in Catia
- Developed sophisticated processes, focusing on efficiency in automated operations including: flow screw, clinch nut, and spot welding processes
- Defined traceability requirements to enhance quality control and accountability
- Achieved significant reductions in manufacturing cycle times by collaborating with tooling suppliers to design custom CNC tooling solutions

Manufacturing Processes

- Welding
- Laser & Torch Cutting
- Special Purpose Machine Development
- Assembly Automation
- CNC Machining
- Sheet Metal Forming

Certifications & Training

- GD&T
- WHIMIS
- Operator Ergonomics
- Business Negotiation
- Lab Safety
- Six Sigma Black Belt
- Engineering Simulations
- CSWA
- Construction Safety
- CQI-9 Heat Treatment
- 8D & 5 Why
- Statistical Process Control
- Advanced MSA
- PFMEA-AIAG
- PFMEA-VDA
- Reliability for Engineering, Production and Quality with MINITAB
- Intro to Driveline Design
- Ethics in Technology
- Power BI
- Statistics Foundations
- Generative AI for Innovation
- Excel Data Analysis
- HTML Essentials
- Python Data Analysis

Experience

Mechanical Lead

University of Alberta EcoCar September 2019 - December 2022

- Led a university team in the design and global competition of hydrogen fuel cell vehicles
- Re-designed the 2023 drivetrain while simultaneously solving overheating issues and improving weight, manufacturability, and efficiency
- Performed topology optimization and FEA simulations using ANSYS to reduce vehicle weight while maintaining safety
- Managed project timelines to ensure timely completion of the vehicle
- Coordinated integration of electrical and fuel cell components
- Helped new members adapt to the team and provided mentorship

Drainage Operations Engineering

EPCOR September 2021 - September 2022

- Oversaw wastewater transport and gained expertise in CORE techniques
- Evaluated costs of current CORE methods against alternatives
- Addressed pump station design issues and optimized parameters within SCADA
- Conducted pump capacity analysis and proposed efficiency improvements
- Evaluated back-up generators in multiple pumping stations and proposed changes which reduced generator operating costs by up to 11%
- Developed and updated technical documents essential to Flow Control Facilities (FCF) such as: Standard Operating Procedures (SOPs); Process Hazard Assessments (PHAs); Hazard and Operability Studies (HAZOPs); and Emergency Response Plans (ERPs)
- Evaluated and interpreted the performance of various flow control facilities, such as RTC gates, pump stations, and stormwater ponds
- Led the design of the trouble truck program, a 24/7 rapid-response unit for urgent wastewater system repairs and maintenance
- Mapped proposed tree planting locations with existing and future FCF facilities to ensure no potential conflicts

Experiential Lab Developer

University of Alberta September 2021 - September 2022

- Developed at-home learning kits consisting of 23 hands-on experiments, instructional videos, and worksheets to facilitate remote solid & fluid mechanics education during the pandemic
- Presented progress in monthly mechanical department meetings and documented progress in multiple project reports
- Managed the supply chain and distribution operations for 300+ remote learning kits, coordinating successful delivery to students across domestic and international locations

Corrosion Technician

Corrosion Service September 2020 - December 2020

- Located TransCanada Pipelines and measured depth for over 600km of pipeline
- Validated depth measurements with the use of a probe
- Responsible for the maintenance and operational function of assigned vehicle and equipment
- Assisted with COR internal audits to ensure compliance
- Completed FLRA safety assessments
- Identified and documented terrain and wildlife hazards in the field